

JUAN F. CASTAÑO

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Summary

Mechanical & Control Engineer and VP of Engineering at Twynable focused on **digital twins, data modeling, and operational analytics** for energy and infrastructure. Experienced shipping **end-to-end solutions** (ingestion → ontology → dashboards → actions) with Python/SQL and cloud services. Strong **systems-thinking**, stakeholder communication, and rapid delivery under ambiguity. .

Core Skills

Programming/Data: Python, SQL, Pandas, NumPy, APIs/JSON, Streamlit, Jupyter

Data & Platforms: Data Modeling, Ontologies, Data Lineage, Ingestion Pipelines, Azure (Data Factory/Storage), CI/CD (basic)

Modeling/Control: Digital Twins, SCADA/PLC, OPC UA, MQTT, PID, FEM/CFD (ANSYS/COMSOL)

Domains: Energy Systems, Grid Optimization, Sensor Telemetry, Reliability/Diagnostics

Soft Skills: Problem Decomposition, Incident Triage, Client Engagement, Clear Communication

Education

B.Sc. Mechanical Engineering

New York University

Aug 2021 – Jun 2025

New York, USA / Abu Dhabi, UAE

- Focus: Energy Systems, Control, Digital Twins, Signal Processing, FEA.

Experience

Vice President of Engineering

Twynable

Aug 2024 – Present

Toronto / Bogotá / Abu Dhabi

- Deployed and scaled digital-twin platforms for infrastructure and power grids, integrating Copernicus/Galileo Earth-Observation with SCADA, IoT, and environmental telemetry.
- Architected multi-source data pipelines (Python, SQL, Azure) for ingestion, transformation, validation; enabled predictive maintenance and operational analytics.
- Modeled complex physical systems into unified data ontologies for real-time monitoring, anomaly detection, and KPI-driven optimization.
- Delivered end-to-end deployments for Alec Engineering, New York University, and the Government of Colombia; implemented secure APIs and decision dashboards.
- Led product strategy & data governance aligned to ISO 23247 / IEC 62832; enforced interoperability, provenance, and system resilience.

Assistant Air Safety Investigator

Air Accident Investigation Sector (AAIS)

Jan 2025 – Sep 2025

Abu Dhabi, UAE

- Applied data-driven diagnostics to reconstruct turbine/airframe behavior via synchronized sensor data (vibration, strain, temperature, oil quality).
- Built automated fault-detection flows (filtering, thresholding) to accelerate anomaly identification; reduced investigation cycle time.
- Mapped sensor outputs to engine subsystems (LP/HP compressors, combustion zones, turbines), enabling granular root-cause analysis and validation of mechanical failure propagation pathways.
- Investigated occurrences under ICAO Annex 13 and GCAA CAR Part VI, including aircraft and UAS incidents, contributed to FDR/CVR analysis and Annex 13-compliant reporting per Doc 9756 Vol. I–III.

Space Commissioning Mentorship, Prof. Simonetta Di Pippo
New York University | European Space Agency ESA

June 2023 – August 2023
Milano, Italy

- Modeled orbital asset behavior using Copernicus Sentinel and Galileo datasets, developing spatiotemporal predictive models for constellation reliability and dual-use (civil/defense) mission analytics.
- Executed commissioning and integration tests for AXIOM Space Station modules and ION Satellite Carriers, ensuring compliance with ESA ECSS and AS9100D quality standards.
- Developed degradation and fault models for sensors under thermal, EMI, and debris stress, deriving MTBF/MTTR metrics and refining Safety-of-Flight thresholds.
- Contributed to AOCS validation for Ariane 6 launch systems, analyzing thruster redundancy and inertial-navigation drift through simulation-based fault injection.

Engineering Researcher (Microfluidics AMMLab)
NYU Abu Dhabi

Oct 2021 – Jun 2025
Abu Dhabi, UAE

- Engineered an automated microfluidic platform for oil quality testing by integrating flow control mechanisms, sensor-actuator feedback loops, and real-time CNN-based diagnostics; currently extending system capabilities through digital twin modeling for predictive maintenance and remote simulation.
- Built sensor-actuator feedback loops; configured PLC/SCADA (OPC UA); drafted troubleshooting notes and acceptance tests used by technicians and assistants, (Fluidic Manipulation).
- Developed oil quality testing device, articulating CNN and Digital Twins for enhanced fluidics diagnoses, publication IEEE: "*Fluidic Manipulation of a Paper Origami for Oil Quality Testing*" (MARSS 2023)

Selected Projects

Saadiyat Island Urban Energy Digital Twin (Python, SQL, Azure, React)

Ingested satellite/weather/SCADA data; designed **ontology** (Asset, Sensor, Event, Region); built transforms, quality checks, and dashboards for **outage risk** and **maintenance planning**.

Sensor Fault Detection Dashboard (Python, Pandas, React)

Near real-time telemetry visualization with anomaly flags; added runbook steps (*detect* → triage → mitigate → RCA → prevent).

Microturbine Optimization Tool (Python, CFD automation, React)

Compared aerodynamic geometries; generated efficiency curves; integrated results into the digital-twin environment for **grid-efficiency what-ifs**, PLC-based control integration, and real-time telemetry testing to ensure mechanical and operational reliability.

Dual Carbon Fiber Microturbine (FEA, CFD, Carbon Fiber, CAD (Solidworks), Arduino, PLC control) Designed and fabricated a dual-mode microturbine with a proprietary airfoil capable of harnessing both drag and lift from traffic-induced vortices, improving localized grid efficiency.

Publications

Synthetic Polymer Based Nano Drug Delivery Systems for Precise Alzheimer's Disease Therapy (SciFiniti 2025)

Fluidic Manipulation of a Paper Origami for Oil Quality Testing (IEEE 2023)

Origami Paper Microfluidic Assay To Test Edible Oil Quality (MicroTAS 2023)

Languages

English (Fluent); Spanish (Fluent); Italian (Fluent); Portuguese (Intermediate); Arabic (Basic)

Engineering Expertise

Forward Deployed Software Engineer; Digital Twin; Ontology; Data Pipeline; Data Modeling; Foundry; Python; SQL; Pandas; APIs; Azure; SCADA; OPC UA; MQTT; Sensors; Telemetry; Incident Triage; Operational Analytics; Energy Systems; Grid Optimization; Reliability